when $i \neq j$, then

$$P\left(\cup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Axiom-3 also holds for a set of finite number of mutually exclusive events. If $A_1, A_2, \ldots,$ An are mutually exclusive events in S and n is a finite positive integer, then

$$P(A_1 \cup A_2 \ldots A_n) = P(A_1) + P(A_2) + \ldots + P(A_n).$$

It may be noted that the previous two approaches to probability satisfy all the above three axioms.

8.3.2 Basic Theorems of Probability

Theorem 8.1: The probability of impossible event is 0 i.e., $P(\phi) = 0$.

Proof: Let $A_1 = S$ and $A_2 = \phi$. Then, A_1 and A_2 are mutually exclusive.

$S = A_1 \cup A_2 = S \cup \phi$

Thus, by Axiom -3,

$$P(S) = P(S) + P(\phi)$$

Since by Axiom-2, $P(S) = 1$,

$$1 = 1 + P(\phi)$$

Hence, $P(\phi) = 0$.

Theorem 8.2: If S is the sample space and A is any event of the experiment, then $P(\bar{A}) = 1 - P(A)$.

Proof:

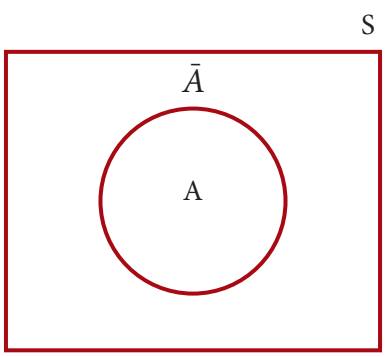


fig. 8.5 Venn diagram

Since A and $\bar{A}$ are mutually exclusive,

$$A \cup \bar{A} = S. \text{ (By Axiom -3)}$$